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- MVA Master 2025/2026 **Deep Learning** Project -
 - Making **Transformers** Solve **Compositional** Tasks -

Extension of the paper by [Ontanon et al., 2022]

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I. Context

Natural language processing (and AI more broadly) has been transformed by **large language models (LLMs)**.

These models achieve SOTA performance in **Out-Of-Distribution (OOD)** generalization.



I. Context

At the heart of LLMs is the **Transformer** architecture and its **attention mechanism**.

Thus, Ontanon et al. (Google Research, 2022) propose to study the OOD generalization capacity of small Transformers trained on **toy synthetic tasks**:

Reverse:

Input: 1 3 3 7 2 [END]
Output: 2 7 3 3 1 [END]

Duplication:

Input: 1 3 5 7 2 [END]
Output: 1 3 5 7 2 1 3 5 7 2 [END]

Addition:

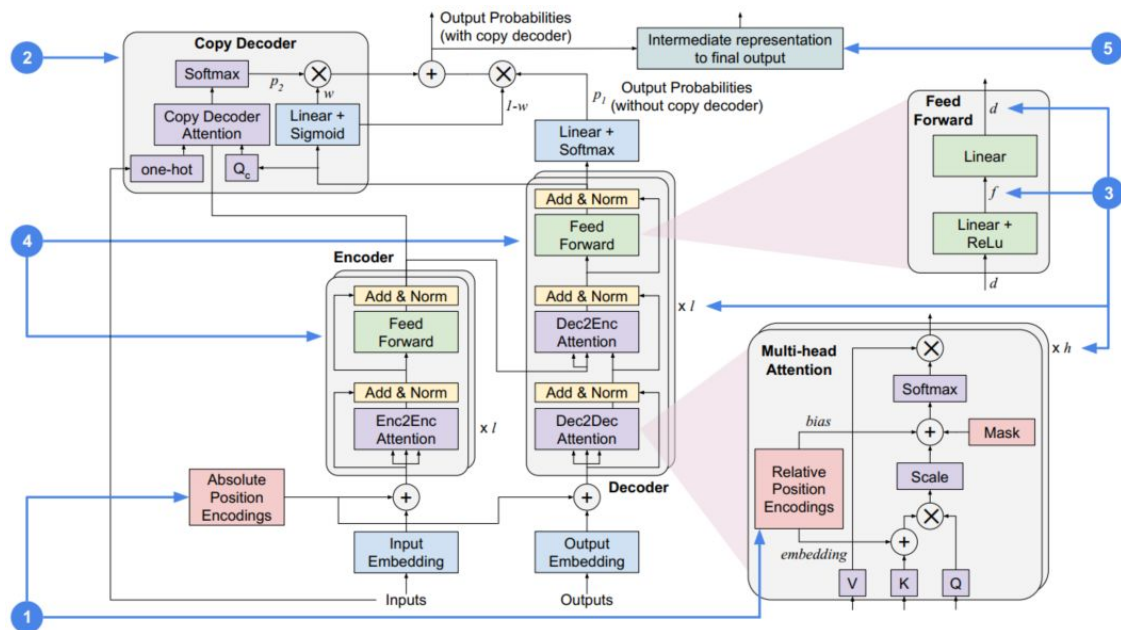
Input: # # # 3 6 7 [SEP] # # 1 4 9 1 [END]
Output: # # 1 8 5 8 [END]

Cartesian:

Input: 1 2 3 [SEP] a b [END]
Output: 1 a [SEP] 2 a [SEP] 3 a [SEP]
1 b [SEP] 2 b [SEP] 3 b [END]

I. Context

Specifically, they analyze the impact of different **architectural choices** on the Transformer OOD generalization.



Excerpt from Ontanon et al. 2022

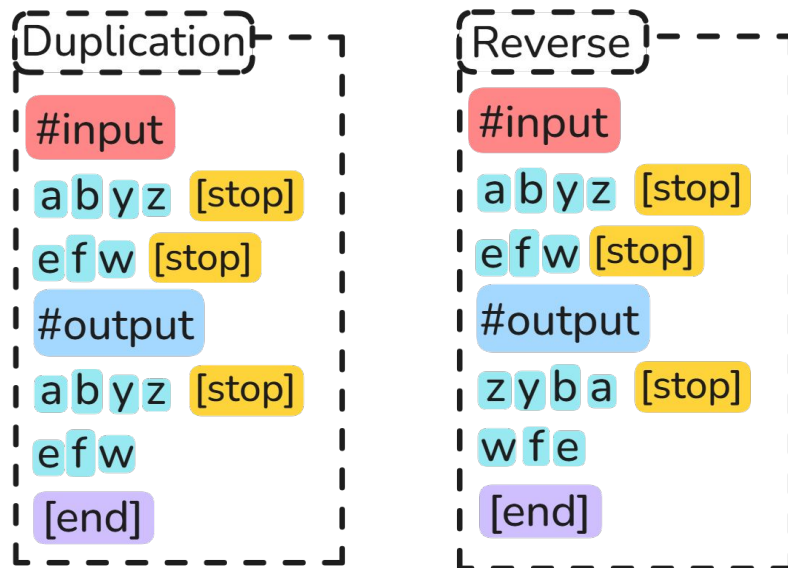
I. Context

Limitations of their study

- ➡ Only studies Encoder-Decoder Transformers, which are actively superseded by Decoder-Only Transformers in LLMs.
- ➡ The OOD generalization tests were conducted by extending the inputs size in the synthetic tasks, but did not consider extending the inputs number.
- ➡ Learnable absolute positional encoding was not considered in the study (only sinusoidal), while the relative positional encodings they considered were learnable embeddings.

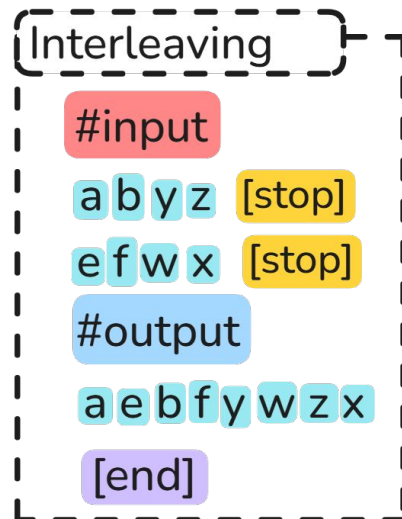
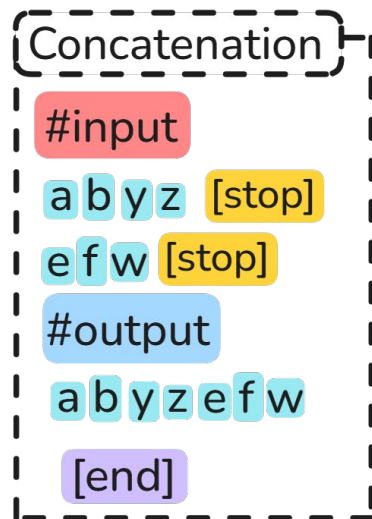
II. Defining Our Synthetic Tasks

We adapt two tasks from Ontanon et al. to our **multi-inputs** setup:



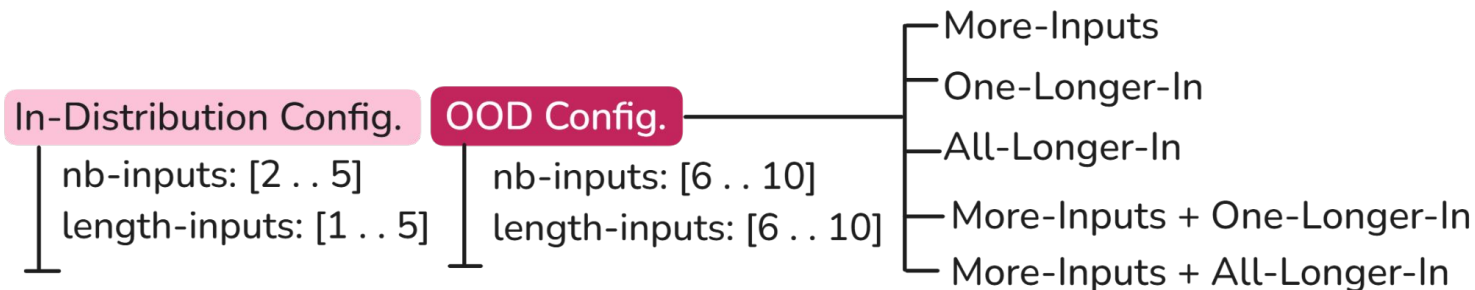
II. Defining Our Synthetic Tasks

We also propose **two new** synthetic tasks that were not considered:



III. Experimental Setup

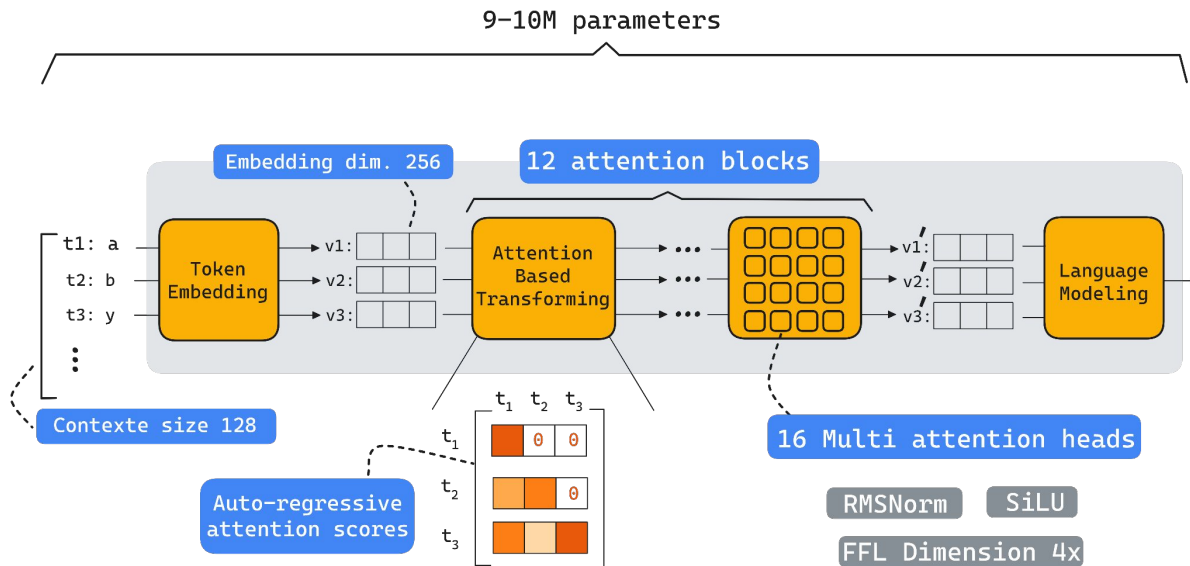
We opt for the following **ID-OOD configuration** for our data:



III. Experimental Setup

We target the decoder-only **GPT architecture**

We **first** identify a **base model** configuration capable of achieving **almost perfect ID-accuracy** i.e. >99% over 10K test samples:



III. Experimental Setup

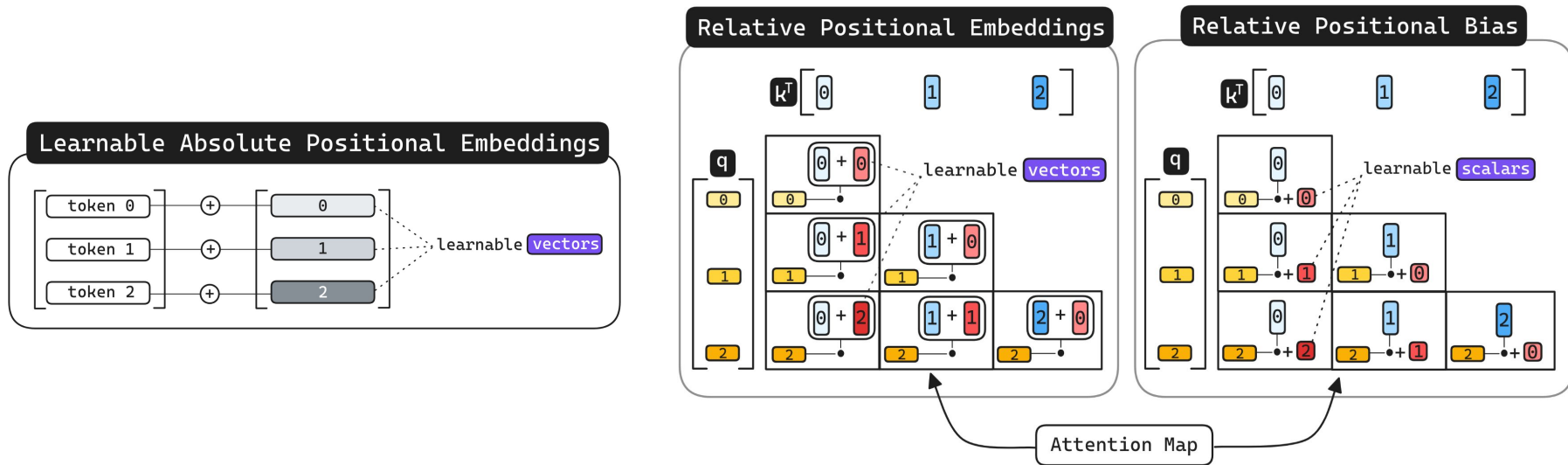
Motivation for almost perfect ID-accuracy

Assessing the OOD capacity shouldn't make sense for an **underperforming** model in the ID regime.

This point was not highlighted in the original paper.

III. Experimental Setup

We compare three different **learnable positional encodings** for out GPT:



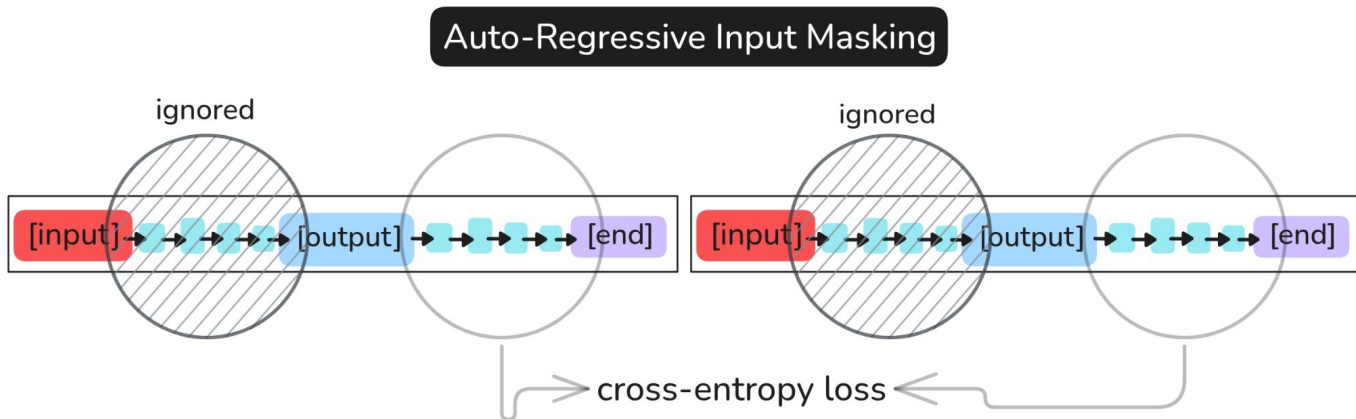
III. Experimental Setup

Beyond architecture: Mimicking Encoder-Decoder with Auto-Regressive Input-Masking

Encoder-Decoder (ED) models have an **inductive bias** that separates input (in encoder) from output (in decoder).

Auto-regressive models lack this capacity → may hinder the generalization.

We propose to test an Auto-Regressive Input-Masking which **mimics ED input-output** separation



IV. Results and Conclusion

Results Analysis:

➤ Despite almost-perfect ID-accuracy for all tasks/models . . . **OOD degradations** had **significant variance** across models, tasks, and OOD-mode difficulty:

Concatenation													More-Inputs + One-Longer-In			More-Inputs + All-Longer-In		
In-Distribution			More-Inputs			One-Longer-In			All-Longer-In				APE RPE RPB			APE RPE RPB		
APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB		APE	RPE	RPB	APE	RPE	RPB
NTP	99.99	100	99.99	60.78	89.91	50.92	92.69	99.92	99.40	40.94	81.48	74.30	33.93	73.32	38.64	0.00	11.34	0.45
	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%

Reversing													More-Inputs + One-Longer-In			More-Inputs + All-Longer-In		
In-Distribution			More-Inputs			One-Longer-In			All-Longer-In				APE RPE RPB			APE RPE RPB		
APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB		APE	RPE	RPB	APE	RPE	RPB
NTP	99.85	100	99.99	0.17	8.04	16.46	6.42	17.71	0.01	0.23	0.69	0.00	0.00	0.33	0.00	0.00	0.00	0.00
	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%

IV. Results and Conclusion

Results Analysis:

Learnable **Absolute** Positional Embeddings could **outperform relative** encodings only **occasionally**:

Interleaving

	Interleaving									More-Inputs +			More-Inputs +					
	In-Distribution			More-Inputs			One-Longer-In			All-Longer-In			One-Longer-In			All-Longer-In		
	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB
without Input-Masking	99.97	100	100	12.56	0.00	0.44				0.00	0.00	0.00				0.04	0.00	0.00
	%	%	%	%	%	%				%	%	%				%	%	%

Duplicating

Duplicating												More-Inputs +			More-Inputs +			
In-Distribution				More-Inputs			One-Longer-In			All-Longer-In			One-Longer-In			All-Longer-In		
APE RPE RPB				APE RPE RPB			APE RPE RPB			APE RPE RPB			APE RPE RPB			APE RPE RPB		
without Input-Masking	99.97	100	99.99	14.36	6.34	16.84	13.60	15.37	0.00	0.94	0.00	0.00	1.86	0.06	0.00	0.00	0.00	0.00
	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%

IV. Results and Conclusion

Results Analysis:

➡ Input-Masking does not degrade ID-accuracy and dramatically improves OOD but only when base OOD accuracy isn't too low, otherwise it may backfire.

Concatenation													More-Inputs + One-Longer-In			More-Inputs + All-Longer-In		
	In-Distribution			More-Inputs			One-Longer-In			All-Longer-In								
	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB
without Input-Masking	99.99 %	100 %	99.99 %	60.78 %	89.91 %	50.92 %	92.69 %	99.92 %	99.40 %	40.94 %	81.48 %	74.30 %	33.93 %	73.32 %	38.64 %	0.00 %	11.34 %	0.45 %
with Input-Masking	100 %	100 %	100 %	75.12 %	98.71 %	97.90 %	99.48 %	99.98 %	99.99 %	49.42 %	98.54 %	85.37 %	47.00 %	97.43 %	92.81 %	0.00 %	86.82 %	2.78 %

Reversing													More-Inputs + One-Longer-In			More-Inputs + All-Longer-In		
	In-Distribution			More-Inputs			One-Longer-In			All-Longer-In								
	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB	APE	RPE	RPB
without Input-Masking	99.85 %	100 %	99.99 %	0.17 %	8.04 %	16.46 %	6.42 %	17.71 %	0.01 %	0.23 %	0.69 %	0.00 %	0.00 %	0.33 %	0.00 %	0.00 %	0.00 %	0.00 %
with Input-Masking	100 %	100 %	100 %	9.92 %	0.00 %	18.40 %	9.20 %	8.52 %	0.00 %	0.03 %	0.23 %	0.00 %	0.70 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %

1

Limited number of positional encoding techniques were tested → explore the many more remaining:

RoPE, ALiBi ...

2

Input-Masking is more supervised → explore architectures that can discover higher level concepts such as input/output in a more self-supervised way:

Hierarchical reasoning models, concept models